VPA A2 Level Computer Science — Semester 2 Course Overview

Victoria Park Academy — A2 Level Computer Science

Semester 2 Course Overview: Final Sprint to A-Level Examinations

Academic Year: 2025 – 2026 | Semester: 2 (January – May) | Duration: 18 Weeks

Exam Board: Cambridge International A-Level Computer Science (9618) | Final Exams: May/June 2026

Focus: Paper 4 Practical Programming Project / NEA-style assessment, intensive revision, mock exam series, final polish.

1. Student Profile

By Semester 2, A2 students have completed AS Level (Paper 1 and Paper 2) and A2 Level Semester 1 (Paper 3 Advanced Theory). They now enter the final phase before A-Level examinations. This semester demands:

Independent project management for the Practical Programming Project (Paper 4 equivalent).

Systematic revision of all A-Level content (AS + A2) across Papers 1 and 2.

Exam technique mastery through repeated mock examinations and targeted feedback.

Resilience and self-regulation to manage workload and stress in the final months.

Entry Competencies (Expected at Start of Semester 2)

Domain	Expected Competency
Theory (Paper 1)	Can explain data representation, hardware, software, communication, security, and AI concepts at AS level; has begun A2 advanced theory (Paper 3).
Programming (Paper 2)	Proficient in Python; can write structured programs using selection, iteration, functions, file handling, and basic OOP.
Problem Solving	Can decompose problems, design algorithms using pseudocode and flowcharts, and trace code execution.
Project Skills	Understands the systems development life cycle (SDLC); has experience with small programming tasks.

Domain	Expected Competency
Exam Technique	Has sat at least two full mock exams; knows command words and mark allocation strategies.

2. Curriculum Map: 18-Week Journey

The Semester 2 curriculum follows a proven cycle: Revise ➞ Mock ➞ Analyse ➞ Target ➞ Revise ➞ Mock ➞ Final Polish.

Phase	Weeks	Focus	Key Activities
Phase 1: Practical Project	1 - 9	Paper 4 Practical Programming Project	Project planning, analysis, design, development, testing, documentation, evaluation
Phase 2: Intensive Revision	10 - 14	Paper 1 & Paper 2 Content Review	Topic-by-topic revision, past paper questions, flipped learning, peer teaching
Phase 3: Mock Series 2	15 - 16	Mock Exams 4 & 5	Timed mock exams under exam conditions, marking, feedback, target setting
Phase 4: Final Polish	17 - 18	Confidence Building & Exam Logistics	Final revision clinics, exam day preparation, wellbeing support

3. Assessment Calendar

□□ Critical: This semester includes 5 full mock examinations. Every mock is a diagnostic opportunity. Students must treat each mock as a real exam.

Mock	Week	Paper	Duration	Marks	Purpose
Mock 2	Week 3	Paper 1 (Theory)	1h 45m	75	Baseline assessment after initial project work; identify theory gaps early
Mock 3	Week 6	Paper 2 (Programming)	2h	75	Programming and problem-solving under timed conditions

Mock	Week	Paper	Duration	Marks	Purpose
Mock 4	Week 15	Paper 1 (Theory)	1h 45m	75	Full A-Level Paper 1 mock; final theory diagnostic
Mock 5	Week 16	Paper 2 (Programming)	2h	75	Full A-Level Paper 2 mock; final programming diagnostic
Project Review	Week 9	Paper 4 Equivalent	Ongoing	75	Formative assessment of practical project; summative mark awarded

4. Standard Lesson Structure

Every lesson in Semester 2 follows a consistent structure to maximise learning efficiency and exam readiness.

Time	Activity	Purpose
0 - 5 min	Starter / Retrieval Practice	Activate prior knowledge; low-stakes quiz or flashcard drill
5 - 10 min	Learning Objective & Success Criteria	Students know exactly what they must achieve by lesson end
10 - 35 min	Main Activity	Input (teacher modelling), guided practice, or independent work (project/revision)
35 - 45 min	Exam-Style Application	Students answer a past paper or mock question under timed conditions
45 - 50 min	Plenary / Exit Ticket	Self-assessment against success criteria; teacher collects evidence of learning

5. EAL Support Checklist

Bilingual glossary provided for all 50+ key terms this semester

Visual diagrams and flowcharts used to explain abstract concepts

Sentence starters provided for extended written answers

Command words (analyse, evaluate, justify, compare) explicitly taught with Chinese translations

Dual-language project briefs available for the Practical Programming Project

Peer translation buddies assigned for complex technical texts

Extra time considered for mock exams where appropriate

Phonetic pronunciation (Pinyin) included for all Chinese terms

6. Differentiation Framework

Tier	Descriptor	Support / Challenge
Bronze	Students working towards Grade C - E	Scaffolded project templates, partially completed code, word banks, one-to-one check-ins, simplified mark schemes, focus on core topics (60% of marks)
Silver	Students targeting Grade B - A	Standard project briefs, guided debugging sessions, peer review, full mark schemes, balanced coverage of all topics
Gold	Students aiming for Grade A*	Open-ended project briefs, complex algorithmic challenges, independent research, teaching others, full past papers, focus on high-tariff questions and evaluation

7. Bilingual Glossary: 50+ Key Terms for Semester 2

English	中文 (Chinese)	Pinyin	Definition
Practical Programming Project	实践编程项目	Shíjiàn biānchéng xiàngmù	A substantial piece of programming work requiring analysis, design, coding, testing, and evaluation.
Systems Development Life Cycle	系统开发生命周期	Xìtǒng kāifā shēngmìng zhōuqī	The structured process of developing an information system: analysis, design, development, testing, implementation, maintenance.

English	中文（Chinese）	Pinyin	Definition
Feasibility Study	可行性研究	Kěxíngxìng yánjiū	An assessment of whether a proposed project is technically, economically, and operationally viable.
Requirements Analysis	需求分析	Xūqiú fēnxī	The process of determining the functional and non-functional requirements of a system.
Data Dictionary	数据字典	Shùjù zìdiǎn	A centralised repository of information about data: names, types, sizes, descriptions, and relationships.
Normalisation	规范化	Guīfàn huà	The process of organising data in a database to reduce redundancy and improve integrity.
Test Plan	测试计划	Cèshì jìhuà	A document describing the scope, approach, resources, and schedule of testing activities.
Normal Data	正常数据	Zhèngcháng shùjù	Test data that is valid and within the expected range.
Boundary Data	边界数据	Biānjiè shùjù	Test data at the extreme limits of valid ranges (e.g., minimum, maximum, just inside, just outside).
Erroneous Data	错误数据	Cuòwù shùjù	Test data that is invalid or outside the expected range to test error handling.
Black-box Testing	黑盒测试	Hēi hé cèshì	Testing without knowledge of internal code; focuses on inputs and outputs.

English	中文（Chinese）	Pinyin	Definition
White-box Testing	白盒测试	Bái hé cèshì	Testing with knowledge of internal code; focuses on path coverage and logic.
Technical Documentation	技术文档	Jìshù wéndàng	Documentation for developers or maintainers: system architecture, code comments, data structures.
User Guide	用户指南	Yònghù zhǐnán	Documentation for end users explaining how to operate the system.
Evaluation	评估	Pínggū	A critical review of the project against objectives, including strengths, limitations, and improvements.
Maintenance	维护	Wéihù	The process of modifying a system after deployment to correct faults, improve performance, or adapt to changes.
Iterative Development	迭代开发	Diédài kāifā	A development approach where the system is built in repeated cycles, refining with each iteration.
Agile Methodology	敏捷方法	Mǐnjié fāngfǎ	A flexible approach to development emphasising collaboration, customer feedback, and rapid iteration.
Version Control	版本控制	Bǎnběn kòngzhì	Tracking and managing changes to code over time (e.g., Git).
Debugging	调试	Tiáoshì	The process of identifying, locating, and removing errors (bugs) from code.

English	中文 (Chinese)	Pinyin	Definition
Trace Table	追踪表	Zhuīzōng biǎo	A table used to manually trace the values of variables through each step of an algorithm.
Stub	存根	Cúngēn	A simplified piece of code used to stand in for a more complex component during testing.
Driver	驱动程序	Qūdòng chéngxù	A test harness that calls a module and passes test data to it.
Mock Exam	模拟考试	Mónǐ kǎoshì	A practice exam under exam conditions to assess readiness and identify gaps.
Retrieval Practice	检索练习	Jiǎnsuǒ liànxí	Learning strategy involving recalling information from memory to strengthen retention.
Spaced Repetition	间隔重复	Jiàngé chóngfù	Reviewing material at increasing intervals to enhance long-term memory.
Interleaving	交叉学习	Jiāochā xuéxí	Mixing different topics or problem types in study sessions to improve discrimination.
Exam Technique	考试技巧	Kǎoshì jìqiǎo	Strategies for maximising marks: time management, command words, structure, checking.
Command Word	指令词	Zhǐlìng cí	The verb in an exam question (e.g., state, explain, evaluate) that dictates the required response.
Time Management	时间管理	Shíjiān guǎnlǐ	Allocating appropriate time to each exam question based on marks available.

English	中文（Chinese）	Pinyin	Definition
Mark Scheme	评分标准	Píngfēn biāozhǔn	The document used by examiners to award marks; students should study these to understand expectations.
Grade Boundary	等级边界	Děngjí biānjiè	The minimum mark required to achieve a particular grade (A*, A, B, C, etc.).
Progression Tracking	进度跟踪	Jìndù gēnzōng	Monitoring student performance over time to identify trends and target intervention.
Target Grade	目标等级	Mùbiāo děngjí	The grade a student is expected to achieve based on prior attainment and potential.
Gap Analysis	差距分析	Chājù fēnxī	Identifying the difference between current performance and target performance.
Intervention	干预	Gānyù	Targeted support provided to close a performance gap.
Revision Clinic	复习诊所	Fùxí zhěnsuǒ	Small-group or one-to-one sessions focused on specific weaknesses.
Past Paper	历年试卷	Lìnián shìjuàn	Previous exam papers used for practice and familiarisation.
Examiner Report	考官报告	Kǎoguān bàogào	Feedback from exam boards on common errors and high-scoring approaches.
Confidence Building	信心建设	Xìnxīn jiànshè	Activities designed to reduce exam anxiety and increase self-efficacy.
Wellbeing	身心健康	Shēnxīn jiànkāng	Physical and mental health support during high-pressure periods.

English	中文 (Chinese)	Pinyin	Definition
Exam Day Logistics	考试日安排	Kǎoshì rì ānpái	Practical arrangements for exam day: equipment, timings, travel, identification.
Post-Mortem	考后分析	Kǎo hòu fēnxī	Detailed review of a mock exam to identify errors and plan improvements.
Flipped Learning	翻转课堂	Fānzhuǎn kètáng	Students study content before class; class time is used for application and discussion.
Peer Teaching	同伴教学	Tóngbàn jiàoxué	Students teaching concepts to each other to reinforce understanding.
Metacognition	元认知	Yuán rènzhī	Thinking about one's own thinking; self-monitoring of learning strategies.
Cognitive Load	认知负荷	Rènzhī fùhè	The amount of mental effort required to process information; managing it is key to effective revision.
Dual Coding	双重编码	Shuāngchóng biānmǎ	Combining words and visuals to enhance memory and understanding.
Elaboration	精细化	Jīngxì huà	Explaining and describing ideas with many details to strengthen memory.
Concrete Example	具体例子	Jùtǐ lìzi	Using specific, real-world instances to illustrate abstract concepts.
Abstract Concept	抽象概念	Chōuxiàng gàiniàn	A theoretical idea without a physical referent (e.g., recursion, encapsulation).

English	中文 (Chinese)	Pinyin	Definition
Scaffolding	支架式教学	Zhījià shì jiàoxué	Temporary support structures that help students reach higher levels of understanding.
Formative Assessment	形成性评估	Xíngchéng xìng pínggū	Ongoing assessment to monitor learning and provide feedback for improvement.
Summative Assessment	总结性评估	Zǒngjié xìng pínggū	Assessment at the end of a period to evaluate overall learning (e.g., final exam).
Valid	有效的	Yǒuxiào de	An assessment that measures what it is intended to measure.
Reliable	可靠的	Kěkào de	An assessment that produces consistent results under consistent conditions.

8. Exam Technique Guide

8.1 Command Words

Command Word	中文	Pinyin	What to Do	Marks
State	陈述	Chénshù	Give a brief, accurate answer. No explanation needed.	1
Describe	描述	Miáoshù	Give a detailed account. Use bullet points or structured sentences.	2 - 3
Explain	解释	Jiěshì	Give reasons or causes. Link cause to effect.	3 - 4
Compare	比较	Bǐjiào	Identify similarities and differences. Use a table or structured format.	4 - 6

Command Word	中文	Pinyin	What to Do	Marks
Analyse	分析	Fēnxī	Break down into parts and examine relationships. Draw conclusions.	6 – 8
Evaluate	评估	Pínggū	Make a judgement based on criteria. Consider both sides. Give a conclusion.	8 – 12
Justify	论证	Lùnzhèng	Give reasons to support a decision or conclusion.	6 – 8
Discuss	讨论	Tǎolùn	Examine different aspects and present a balanced argument.	8 – 12

8.2 Time Management Strategy

Paper 1 (1h 45m = 105 minutes, 75 marks): 1.4 minutes per mark.

Paper 2 (2h = 120 minutes, 75 marks): 1.6 minutes per mark.

Strategy:

1. Scan the whole paper first. Identify high-mark questions.
2. Answer all low-tariff questions first to secure marks quickly.
3. Allocate time proportionally: a 12-mark question deserves 16 – 17 minutes.
4. Leave 5 minutes at the end for checking and correcting silly errors.
5. If stuck, move on. Return later with fresh eyes.

8.3 Structure for Extended Answers

For 6+ mark questions, use this structure:

Introduction: Define key terms and state your position.

Point 1 + Evidence: Make a claim, support with technical detail or example.

Point 2 + Evidence: Second claim with supporting evidence.

Counter-argument (for Evaluate/Discuss): Acknowledge an opposing view.

Conclusion: Summarise and make a final judgement.

9. Resources and References

Cambridge International A-Level Computer Science (9618) Syllabus (2024 – 2026)

Cambridge Past Papers (2018 – 2025) — Papers 1, 2, 3, and 4

Cambridge Examiner Reports (2018–2025)

Python 3.12 Documentation (docs.python.org)

Git Version Control Guide (git-scm.com)

VS Code Editor (code.visualstudio.com)

VPA Bilingual Glossary (150 Terms) — Supporting Documents

VPA A2 Semester 2 Assessment Pack — Mock exams, mark schemes, grade boundaries

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Next Review: June 2026 (post-exam analysis)